

VLANs

Advantages of VLANs

- Solve broadcast problem
- Reduce the size of broadcast domains
- Allow us to add an additional layer of security
- Make device management easier
- Allow us to implement the logical grouping of devices instead of location

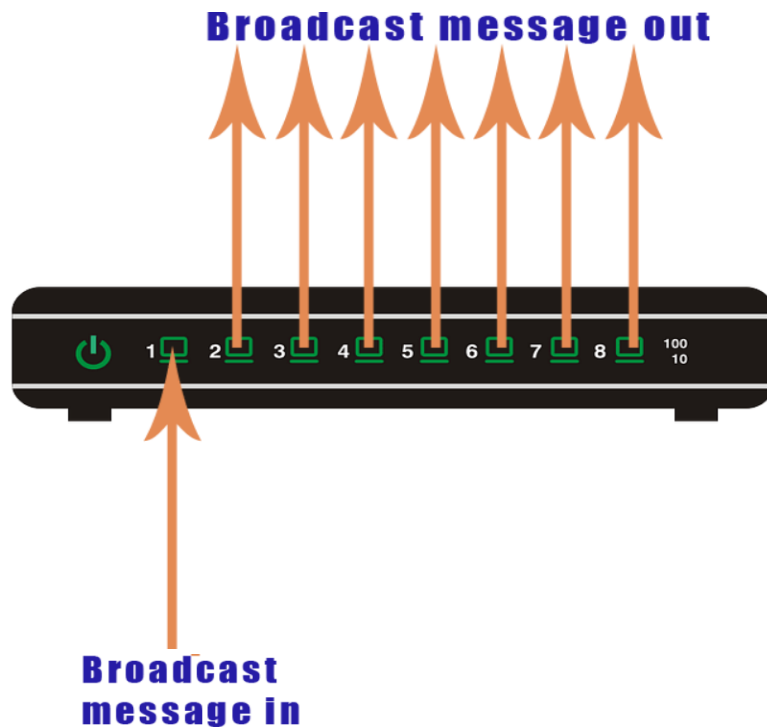
VLANs

Advantages of VLANs

- Solve the broadcast problem (decrease the network performance)
 - switch create a single broadcast domain
 - Use VLAN (unique ID, VLAN ID)
-
- 2. Reduce the size of broadcast domain
 - 3. additional layer of security
 - 4. Make device management easier

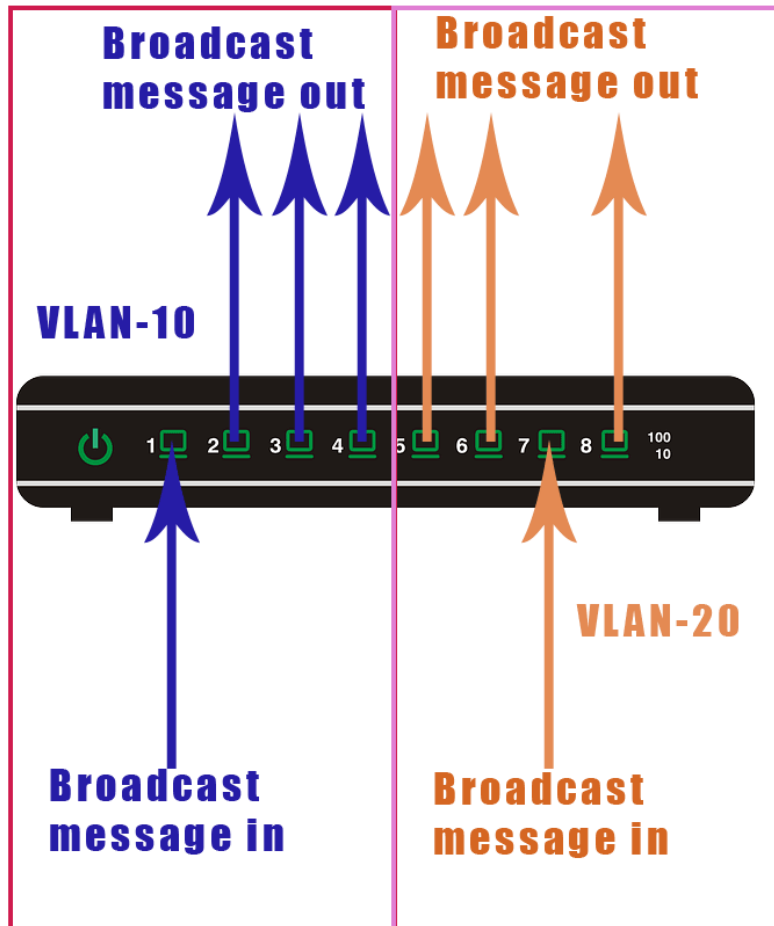
VLANs

- Solve broadcast problem
 - The following image shows how a switch forwards a broadcast message with the default configuration



VLANs

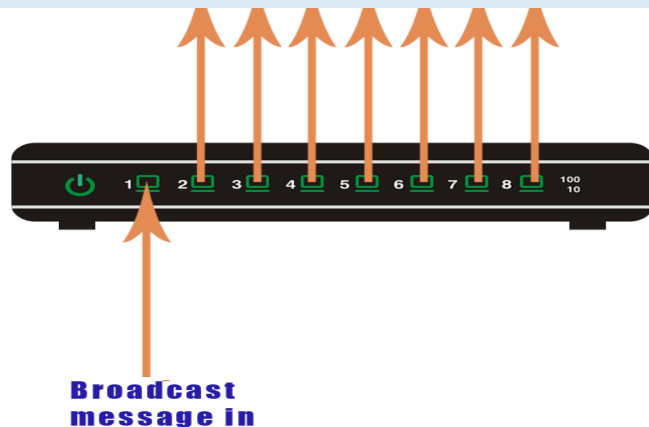
- A switch forwards broadcast messages when it has two VLANs



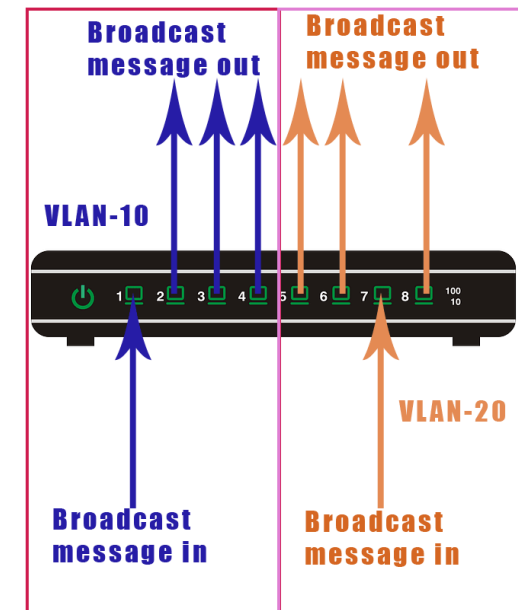
VLANs

- Reduce the size of broadcast domains

- Eg: A network has 10 devices.
- With the default configuration, it has a single broadcast domain that contains 10 devices.



- If create 2 VLANs and assign 5 devices to each VLAN,
- we will have two broadcast domains with five devices in each.
- More VLAN means more broadcast domains with fewer devices.



VLANs

- Allow us to add an additional layer of security
- VLANs enhance network security.
- In a typical layer-2 network,
 - users can access any network resources located on that specific network and users could join a workgroup.
- It could create security issues.
- With VLANs, we can control the users from gaining unwanted access to the resources.
- we can put the group of users that need high-level security into their own VLAN so that users outside of VLAN can't communicate with them.

VLANs

- Make device management easier
- VLANs are a logical approach, a device can be located anywhere in the switched network and still belong to the same broadcast domain.
- We can move a user from one switch to another switch in the same network while keeping his original VLAN.
- For example, a company has a five-story building and a single-layer-2 network.
- In this scenario, VLAN allows us to move the users from one floor to another floor while keeping their original VLAN ID.

Static Route

- Preconfigure routes
- Do not add overhead in the form of routing protocols, advertisements, and learning networks
- Router will not adjust to network changes
- Smaller networks with only several routers, or networks where security is a top priority, often utilize static routers.

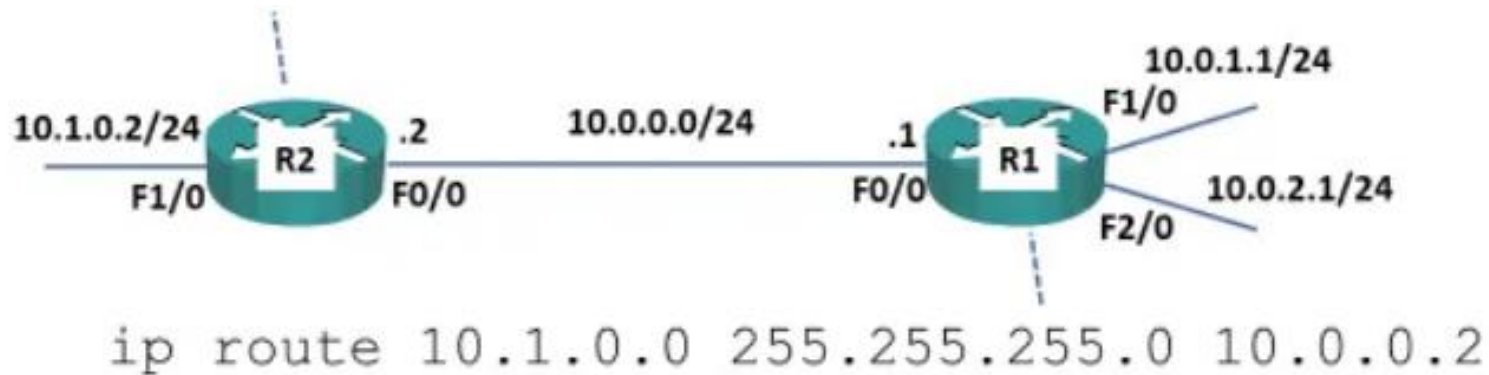
Static Route

- If a router receives traffic for a network which it is not directly attached to, it needs to know how to get there in order to forward the traffic
- An administrator can manually add a static route to the destination, or the router can learn it via a routing protocol

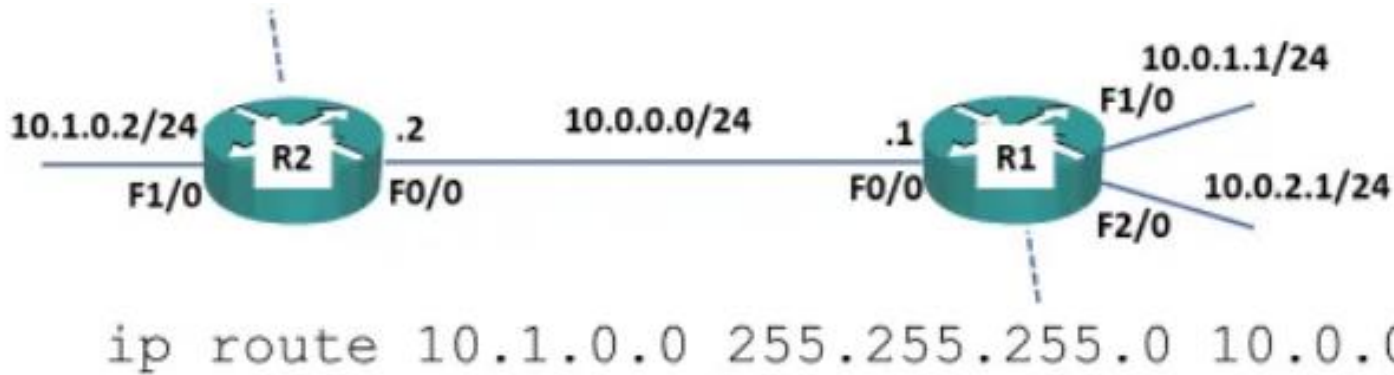


Static Route

- There is another network, 10.1.0.0/24, which is behind the router R2 and R1 doesn't know how to get there yet.
- Therefore, a static route is needed to be added into the R1 routing table in order to reach the said network.



Static Route



R1's route to get to the 10.1.0.0/24 network. The command needed to be added to R1 is:

R1(config)#ip route 10.1.0.0 255.255.255.0 10.0.0.2

It is the subnet it needs to get to, its subnet mask, then the next hop IP address that I need to send traffic to.

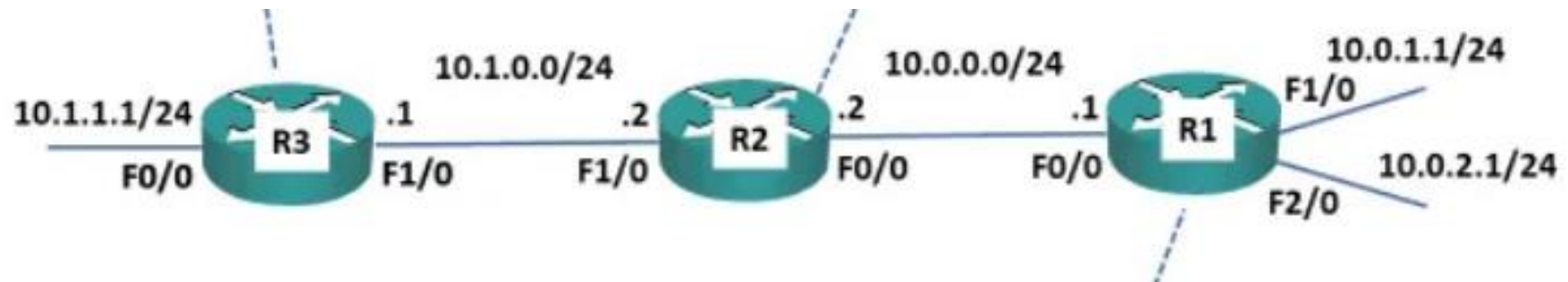
Static Route



- The R2 would need two routes, a route to 10.0.1.0/24 and to 10.0.2.0/24 networks. The commands are:
 - R2(config)#ip route 10.0.1.0 255.255.255.0 10.0.0.1
 - R2(config)#ip route 10.0.2.0 255.255.255.0 10.0.0.1

Static Route

- add another router, R3, to the previous topology
- R3 has got routes to 10.1.1.0/24 and to 10.1.0.0/24 networks which are directly connected to it.
- Right now, it doesn't know how to get to the network 10.0.0.0/24 behind R2, and to the networks 10.0.1.0/24 and 10.0.2.0/24 behind R1.

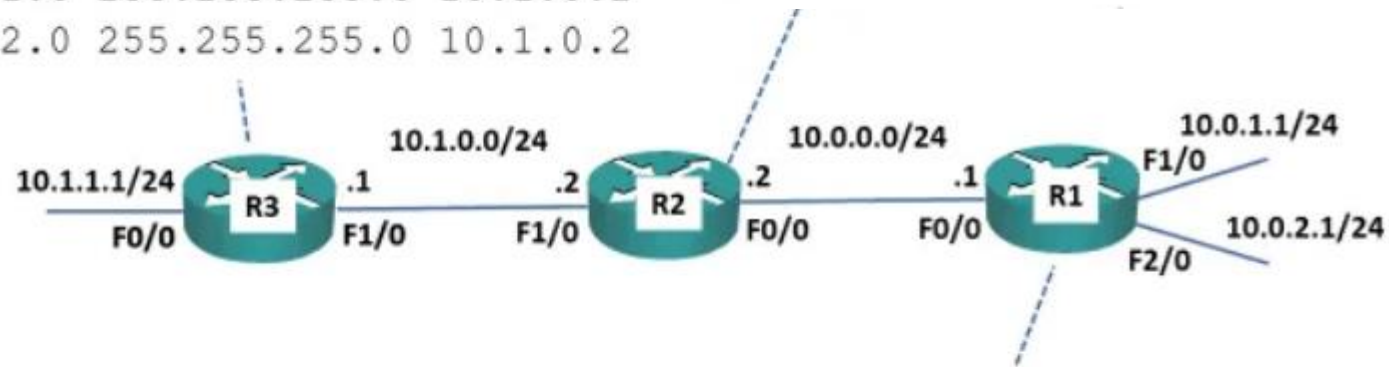


Static Route

- add another router, R3, to the previous topology
- R3 has got routes to 10.1.1.0/24 and to 10.1.0.0/24 networks which are directly connected to it. Right now, it doesn't know how to get to the network 10.0.0.0/24 behind R2, and to the networks 10.0.1.0/24 and 10.0.2.0/24 behind R1.

```
ip route 10.0.0.0 255.255.255.0 10.1.0.2
ip route 10.0.1.0 255.255.255.0 10.1.0.2
ip route 10.0.2.0 255.255.255.0 10.1.0.2
```

```
ip route 10.1.1.0 255.255.255.0 10.1.0.1
ip route 10.0.1.0 255.255.255.0 10.0.0.1
ip route 10.0.2.0 255.255.255.0 10.0.0.1
```



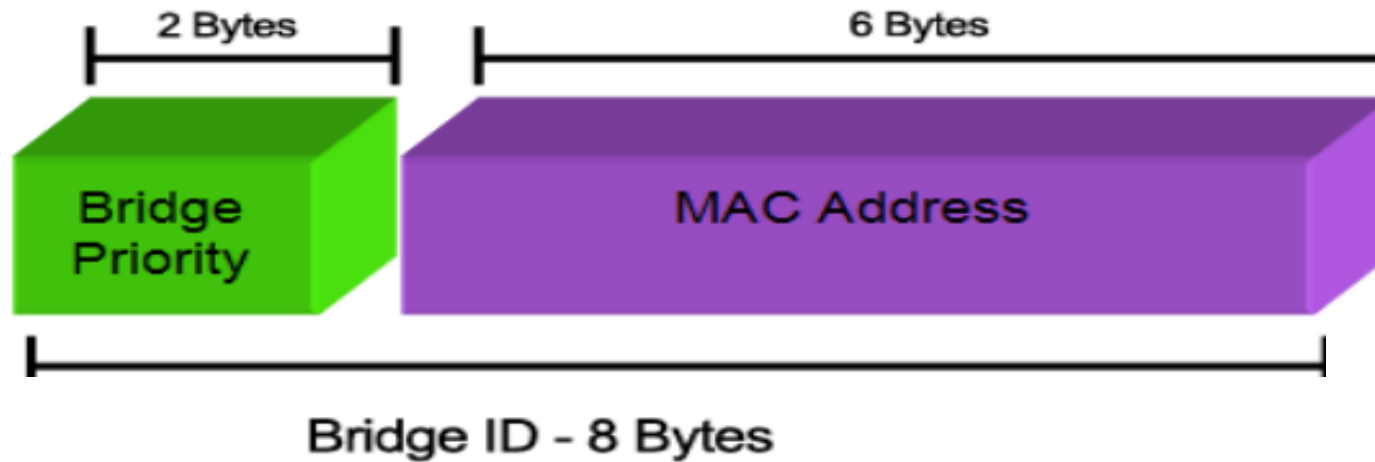
```
ip route 10.1.0.0 255.255.255.0 10.0.0.2
ip route 10.1.1.0 255.255.255.0 10.0.0.2
```

Spanning Tree Protocol

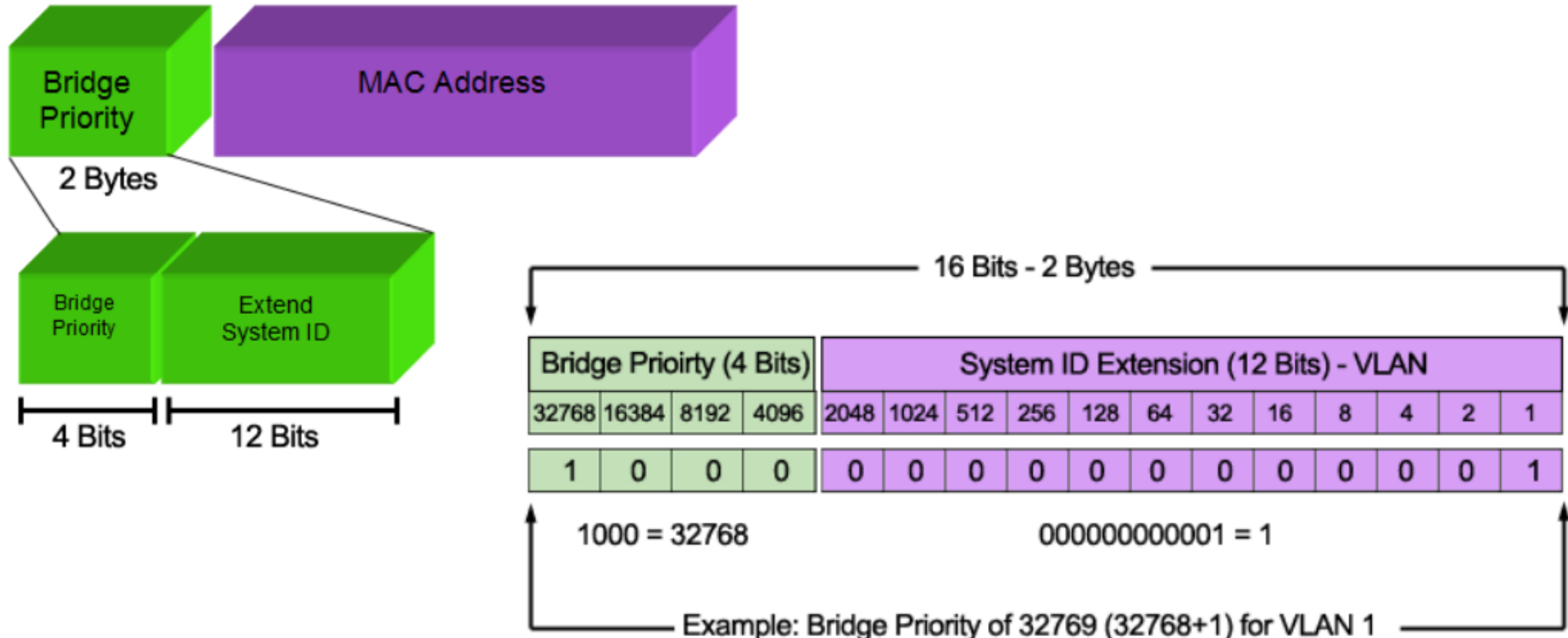
- STP is a Layer 2 network protocol
- STP monitors all network links, identifies redundant connections and disables the ports that can lead to looping
- Control the flow of traffic and prevent looping when redundant paths exist
 - Uses STA algorithm
 - exchange the BPDUs and select the best data path

STP Root Bridge Selection

- Bridge ID (BID) (Bridge Priority + MAC)

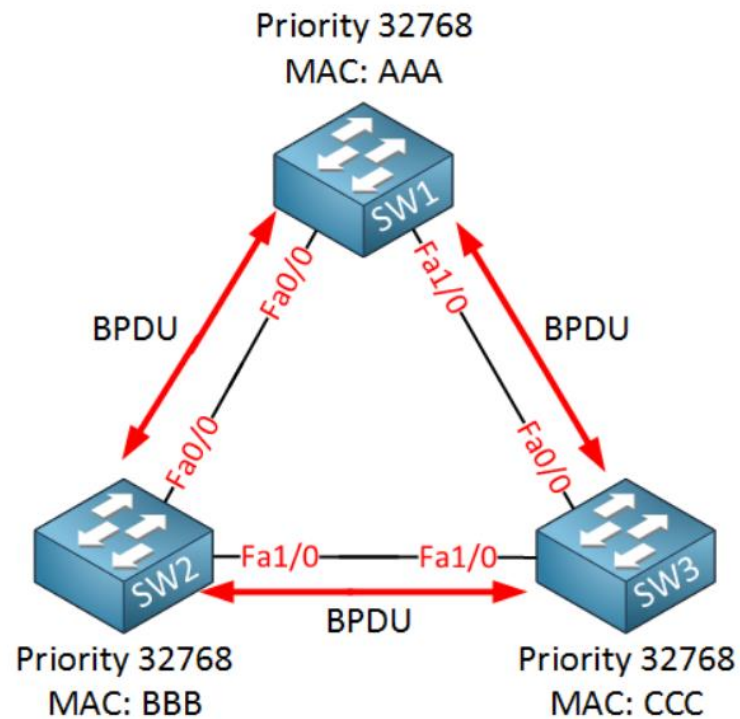


STP Root Bridge Selection



STP Root Bridge Selection

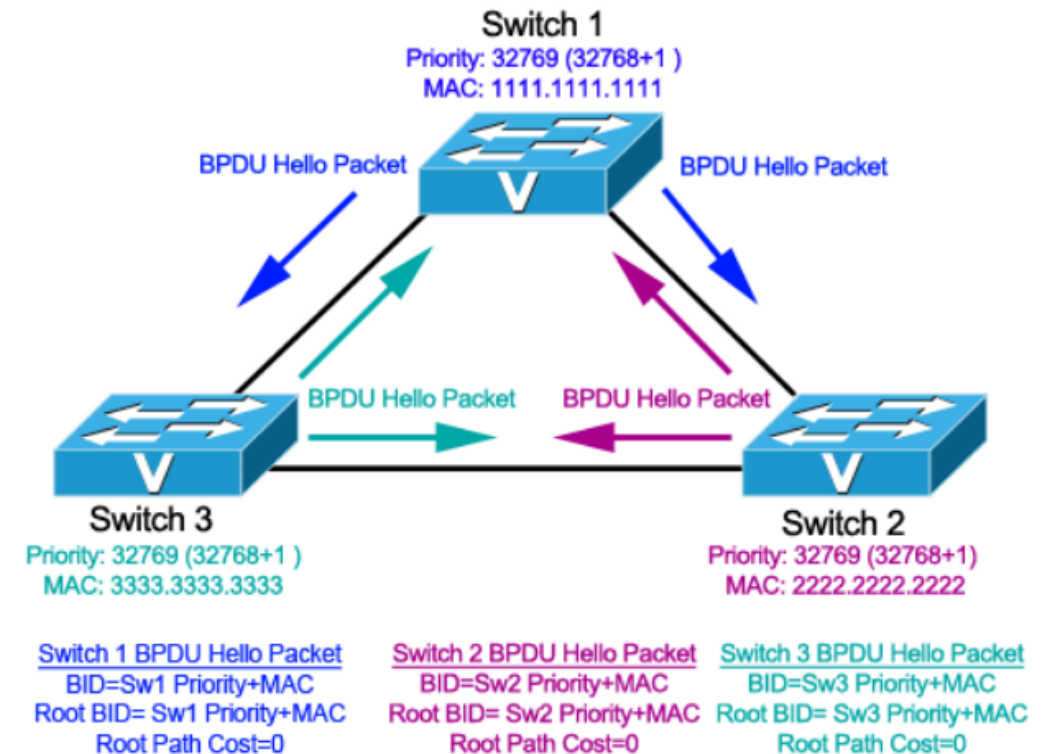
- BPDU is sent between switches as shown below.



STP Root Bridge Selection

- **Switch 1 (SW1).** Has a priority of 32769 and MAC address of 1111.1111.1111. So **its BID becomes 32769.1111.1111.1111**. When SW1 creates its own BPDU, it sets both BID and Root BID to 32769.1111.1111.1111.
- **Switch 2 (SW2).** Has a priority of 32769 and MAC address of 2222.2222.2222. So **its BID becomes 32769.2222.2222.2222**. When SW2 creates its own BPDU, it sets both BID and Root BID to 32769.2222.2222.2222.
- **Switch 3 (SW3).** Has a priority of 32769 and MAC address of 3333.3333.3333. So **its BID becomes 32769.3333.3333.3333**. When SW3 creates its own BPDU, it sets both BID and Root BID to 32769.3333.3333.3333.

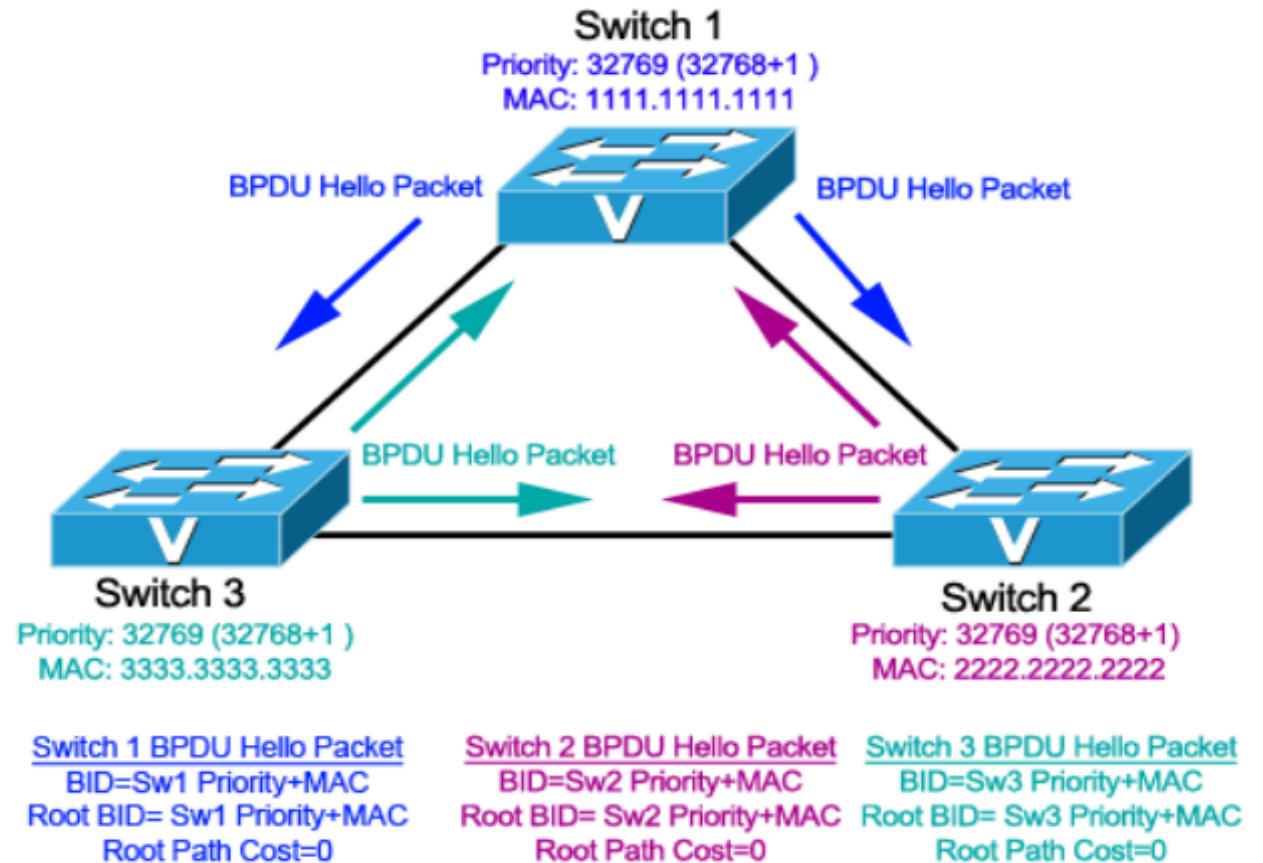
Example



STP Root Bridge Selection

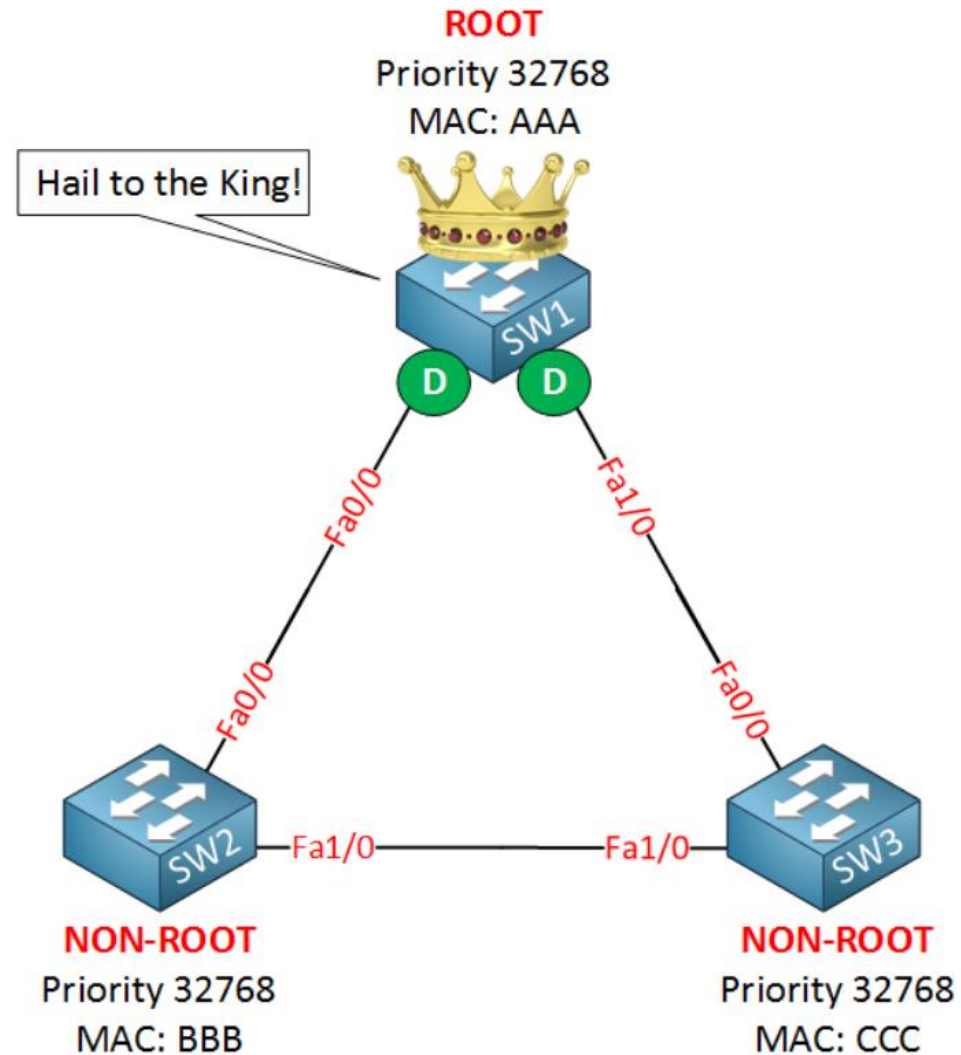
- **SW1** has the lowest **BID** address **Root Bridge** of the network.
- Both **SW2**, and **SW3** agree that **SW1** is **Root Bridge**,
 - and start organizing their respective links into **Root Ports** and **Designated Ports**.

Example



STP Root Bridge Selection

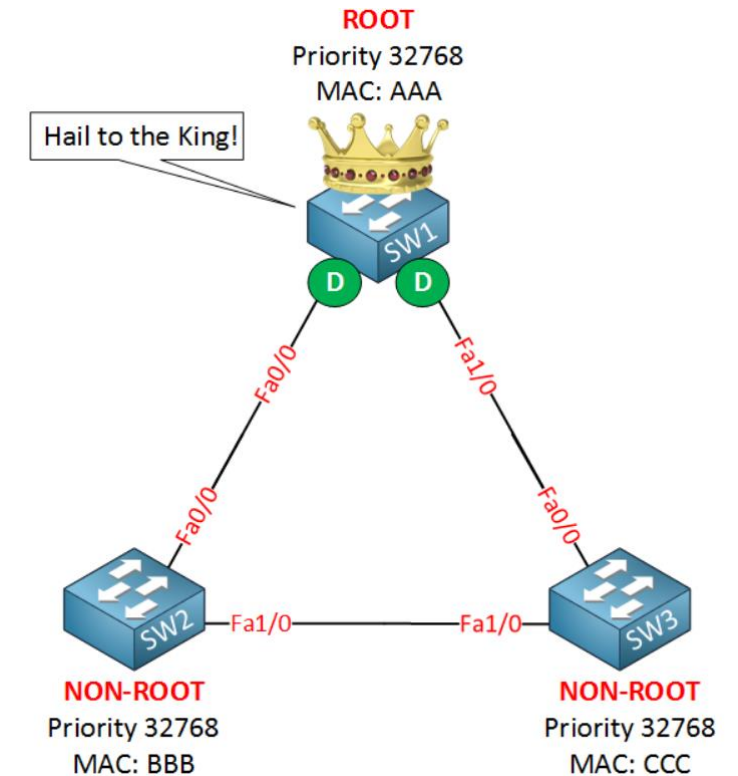
who will become the root bridge?



STP Root Bridge Selection

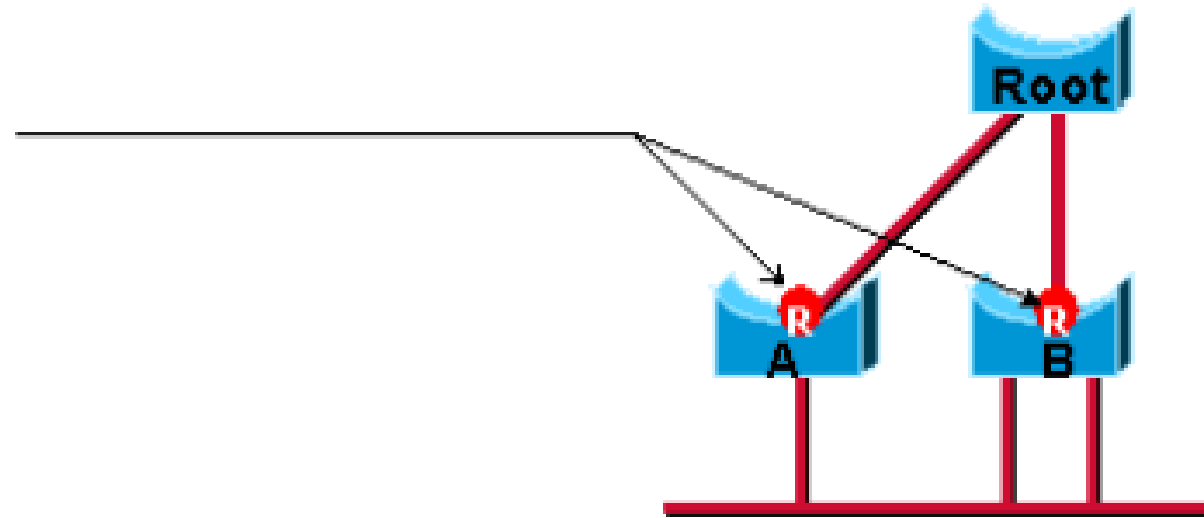
Lowest MAC address

- 00:00:00:00:00:01 is smaller than 00:00:00:00:00:02
- AB:00:00:00:00:00 is smaller than AC:00:00:00:00:00
- AB:CD:EF:12:34:56 is larger than AB:CD:EF:12:34:11
- 00:00:00:55:55:55 is larger than 00:00:00:44:55:55
- FF:FF:FF:FF:FF:FF is larger than 00:00:00:00:00:00



STP Root Port Selection

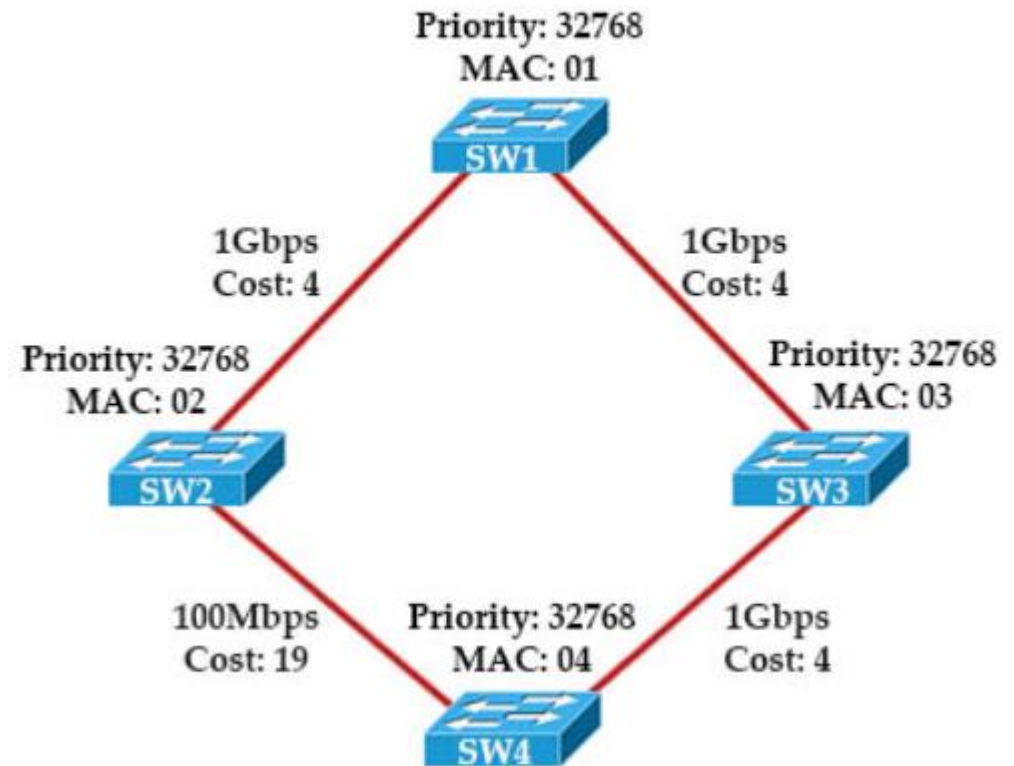
Root Port



- best cost path (shortest path) to root bridge
- a port cost is calculated by evaluating the bandwidth of each link
- If the path cost is the same, the switch will select the port with the lowest sender BID as the selected root port.

STP Root Port Selection

- Lowest accumulated cost on interfaces towards Root Bridge
- Lowest Sender Bridge ID



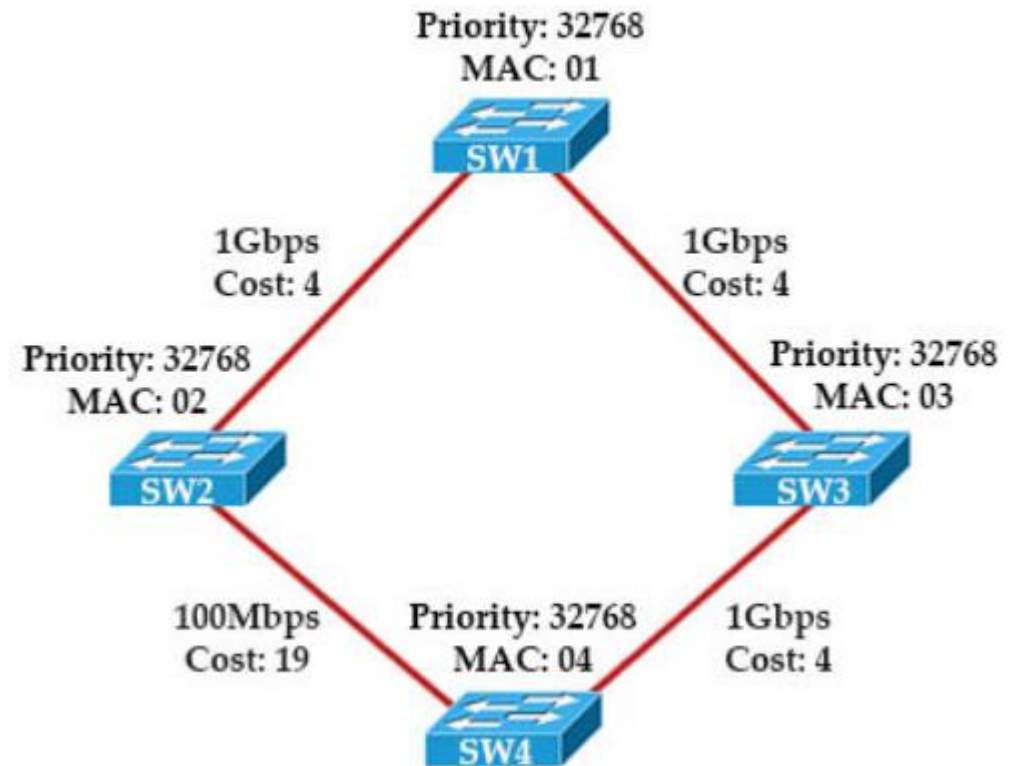
STP Root Port Selection

1. SW2 has two ways to reach SW1

- SW2 - SW1: path cost is $0 + 4 = 4$
- SW2 - SW4 - SW3 - SW1: path cost is $0 + 4 + 4 + 19 = 27$

The path cost is calculated as follows:

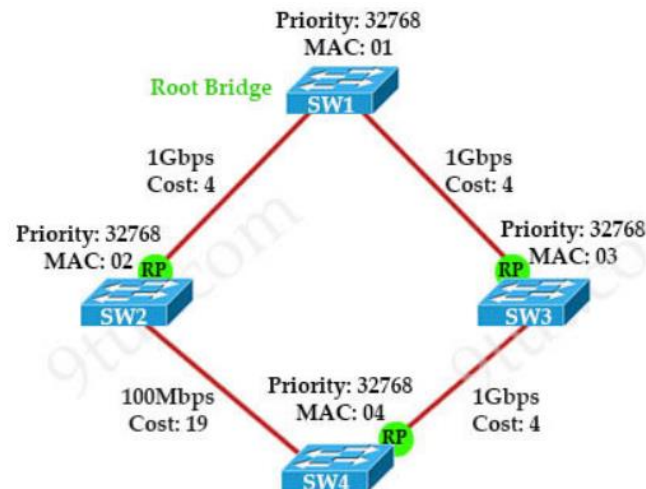
- 10 Mbps = 100
- 100 Mbps = 19
- 1 Gbps = 4
- 10 Gbps = 2



STP Root Port Selection

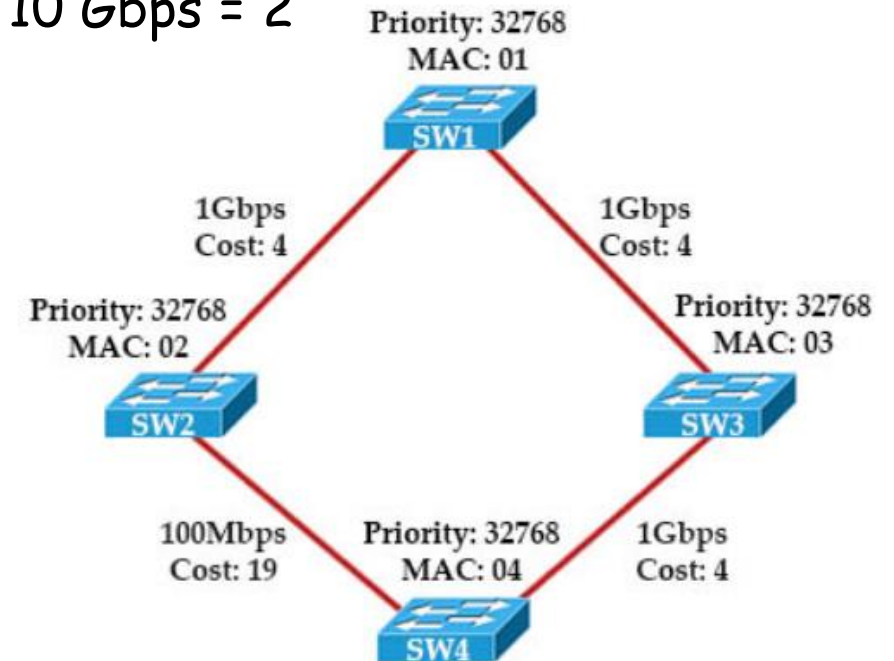
SW4 has two ways to reach SW1:

- SW4 - SW2 - SW1: path cost is $0 + 4 + 19 = 23$
- SW4 - SW3 - SW1: path cost is $0 + 4 + 4 = 8$



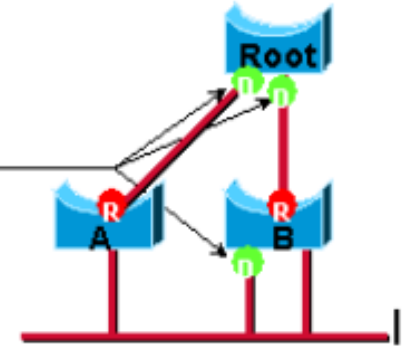
The path cost is calculated as follows:

- 10 Mbps = 100
- 100 Mbps = 19
- 1 Gbps = 4
- 10 Gbps = 2



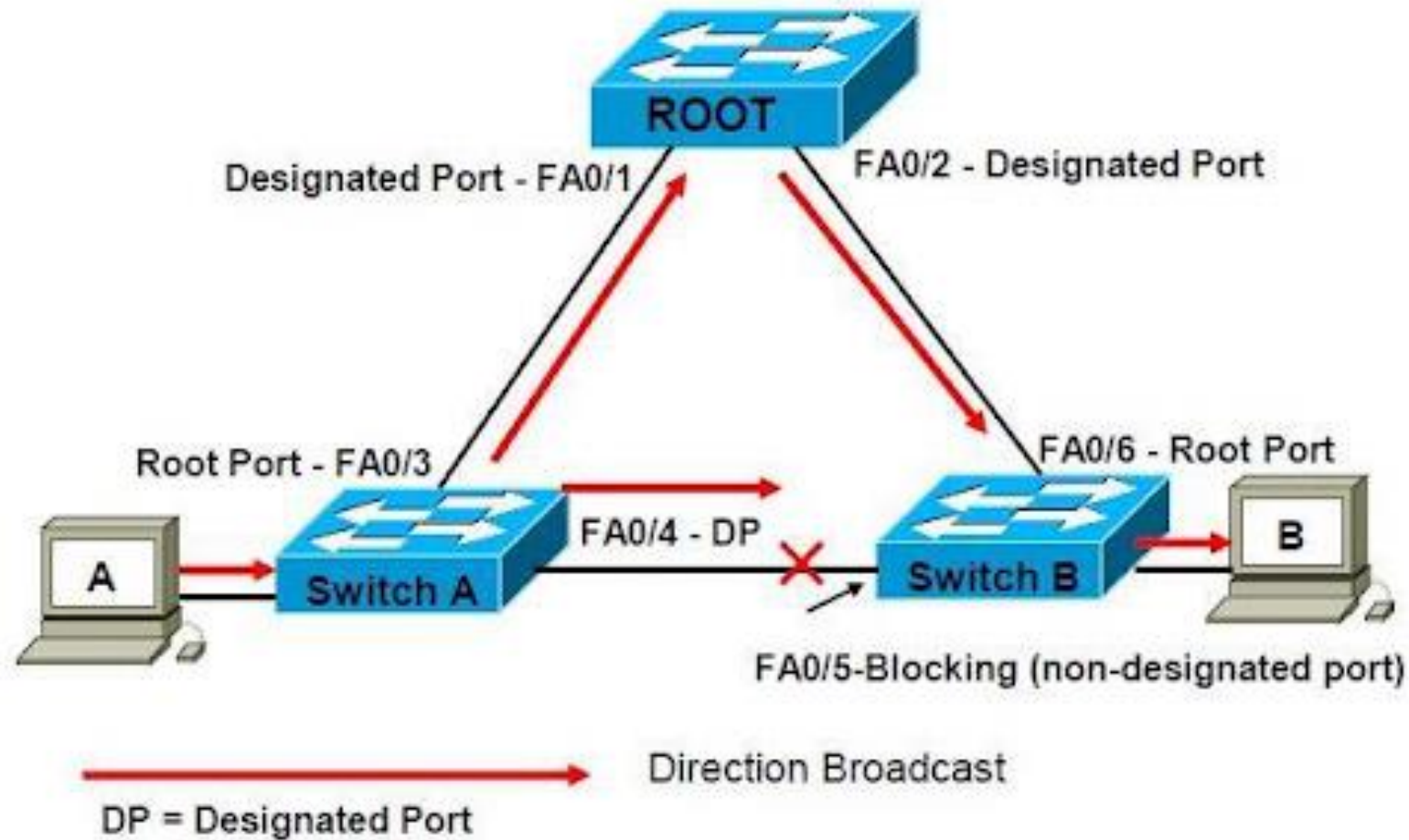
STP designed Port Selection

Designated Port



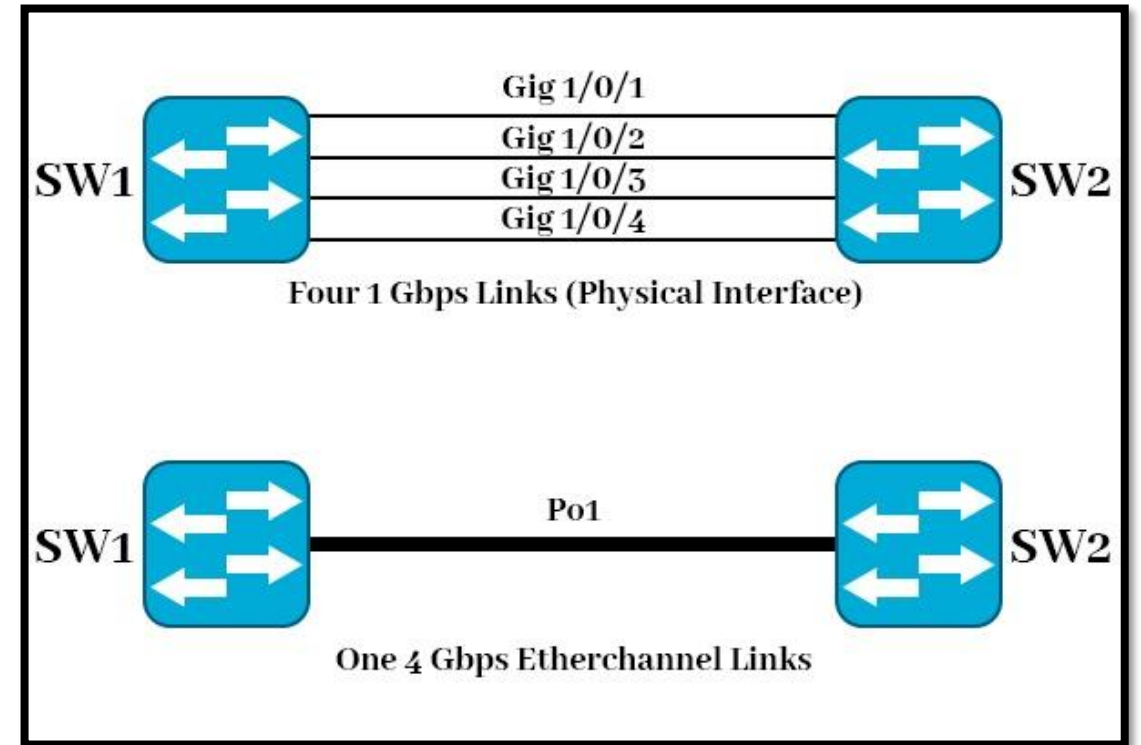
- If there are two paths with equal cost to the root bridge, STP uses the following criteria for best path determination and consequently for determination of designated and non-designated ports on the segment:
 - Lowest root BID
 - Lowest root path cost to root bridge
 - Lowest sender BID

Spanning Tree Protocol



EtherChannel

- it is a port link aggregation
- It allows grouping of several physical Ethernet links to create one logical Ethernet link
- create higher bandwidth links as the traffic needs of the network increase



Generations of Ethernet

Ethernet Technology	Segment Bandwidth
Ethernet	10 Mbps
Fast Ethernet	100 Mbps
Gigabit Ethernet	1 Gbps
10-Gigabit Ethernet	10 Gbps
40-Gigabit Ethernet	40 Gbps
100-Gigabit Ethernet	100 Gbps

EtherChannel feature

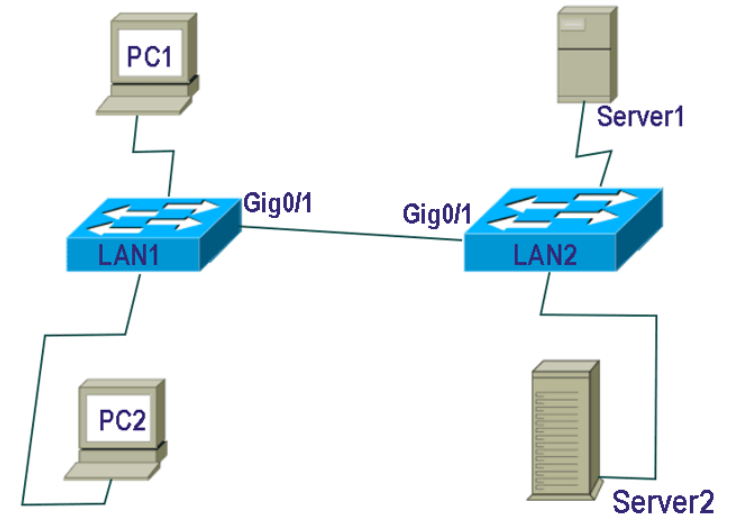
- Provide redundancy
- Provides automatic recovery for the loss of a link by redistributing the load across the remaining links
- Configure all physical ports in an EtherChannel manually
- Each EtherChannel can consists of up to four compatibly configured physical ports.
- Configure all physical ports in an EtherChannel to work at the same criteria such as speed, duplex, and trunk mode.
- Each physical port can be a member of one EtherChannel group.

EtherChannel

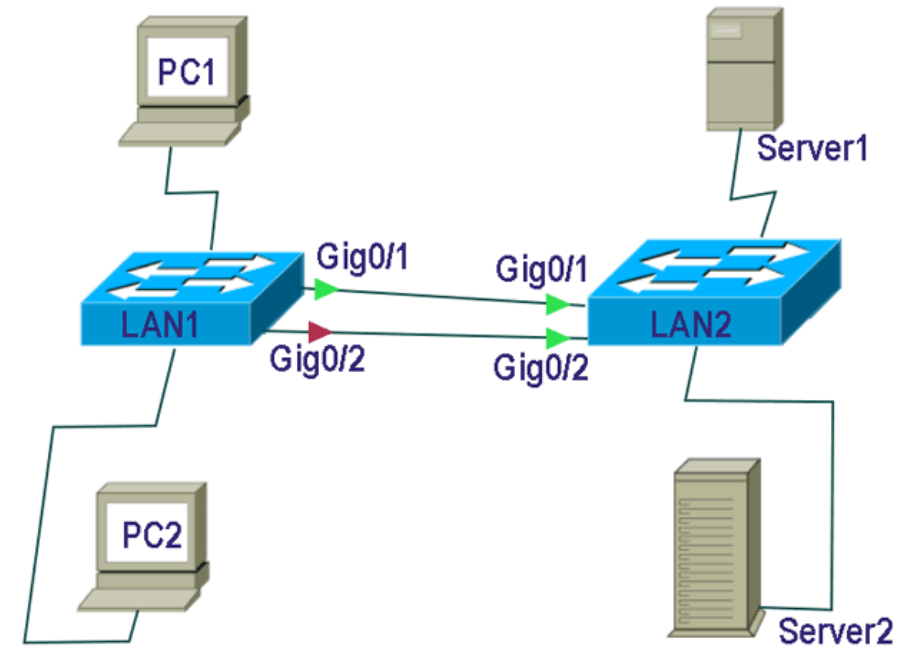
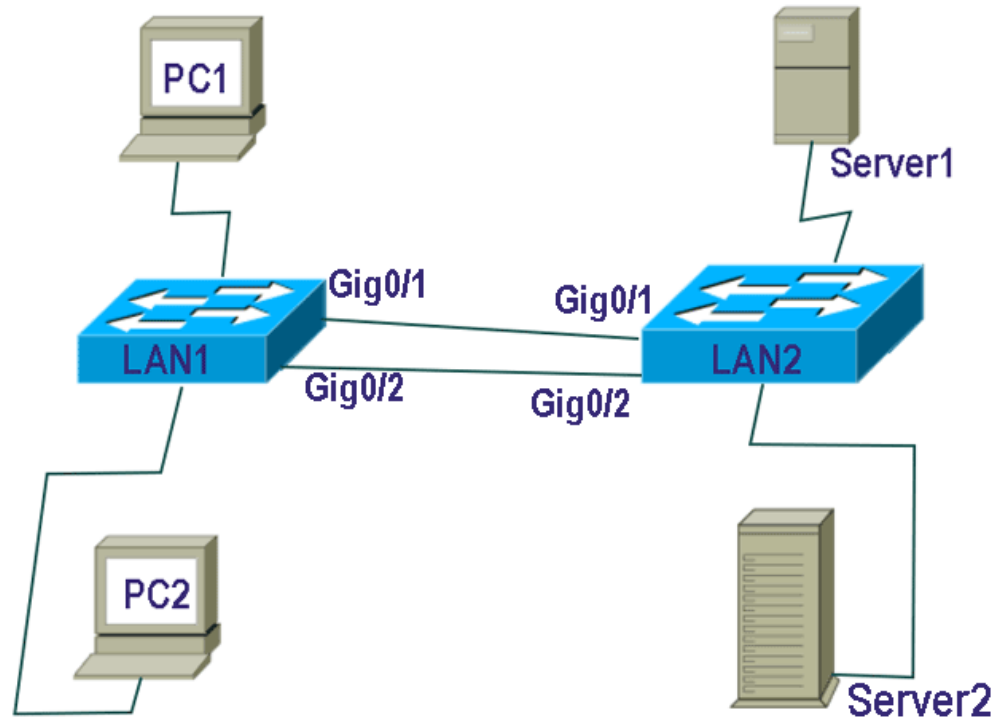
- a port link aggregation
- grouping of several physical Ethernet links to create one logical Ethernet link
- create higher bandwidth links as the traffic needs of the network increase
- higher bandwidth and provide redundancy

EtherChannel

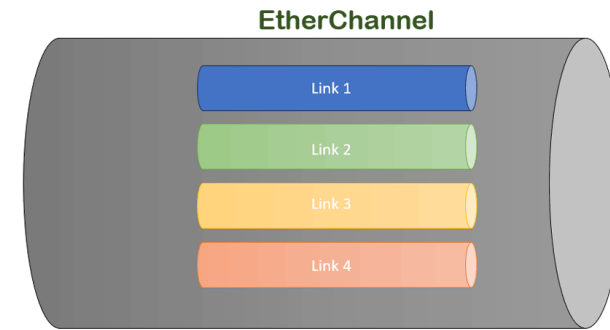
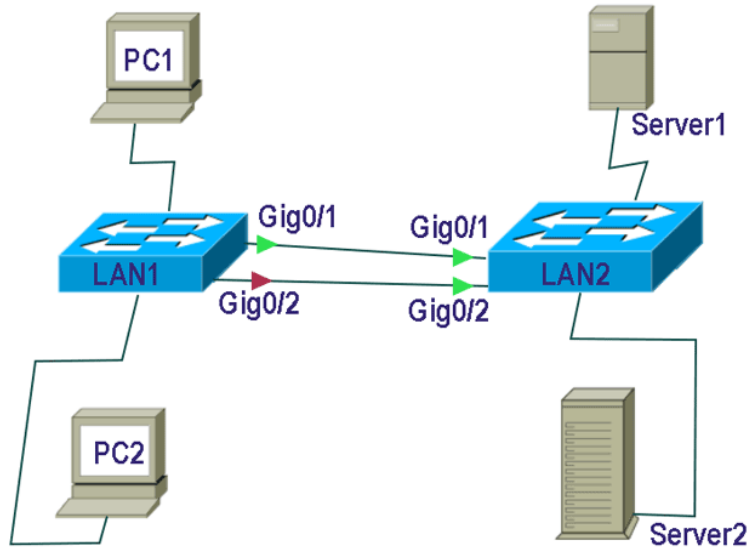
- PC1 generates 800MB of traffic for Server1.
- PC2 generates 600MB of traffic for Server2.
- Both use the same link to reach their destinations.
- The link's bandwidth is 1GBPS.
- It is less than the total traffic sent by both PCs.
- $1 \text{ GBPS (Link's bandwidth)} < 1400 \text{ MB}$
 $(800\text{MB [PC1's traffic]} + 600 \text{ [PC2's traffic]})$



EtherChannel



EtherChannel



EtherChannel V/s Port Channel V/s Channel Group

```
Switch0
Physical Config CLI Attributes
IOS Command Line Interface

Switch>enable
Switch#configure terminal
Enter configuration commands, one per line.
Switch(config)#interface gi0/1
Switch(config-if)#channel-group 1 mode on
Switch(config-if)#exit
Switch(config)#interface gi0/2
Switch(config-if)#channel-group 1 mode on
Switch(config-if)#exit
Switch(config)#exit
Switch#show etherchannel
                        Channel-group listing:
                        -----

Group: 1
-----
Group state = L2
Ports: 2 Maxports = 8
Port-channels: 1 Max Port-channels = 1
Protocol: -
Switch#
```

Type of EtherChannel

1. Link Aggregation Control Protocol (LACP)

- Works with all vendor switches

2. Port Aggregation Protocol (PAgP)

- Works only on Cisco switches

EtherChannel Restrictions/Requirements

- Port Speed
- Port Duplex mode
- Access or trunking state (all must be either access or trunks)
 - If it is an access port, the configured VLAN on the access port
 - If it is a trunk port, the allowed VLANs on the trunk link

EtherChannel

- STP is the default on all Cisco switches.
- It automatically finds loops and block ports that cause the loops.



EtherChannel

- Verify STP port states
 - show spanning-tree command displays the STP port states

```
Switch#show spanning-tree
VLAN0001
  Spanning tree enabled protocol ieee
  Root ID    Priority    32769
             Address     0060.4765.73AE
             Cost        19
             Port        1(FastEthernet0/1)
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    32769 (priority 32768 sys-id-ext 1)
             Address     00E0.8F8A.734B
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time  20
```

Interface	Role	Sts	Cost	Prio.Nbr	Type
Fa0/1	Root	FWD	19	128.1	P2p
Fa0/2	Altn	BLK	19	128.2	P2p
Fa0/3	Altn	BLK	19	128.3	P2p
Fa0/4	Altn	BLK	19	128.4	P2p

STP blocked these ports to remove the loop.

EtherChannel

Switch(config-if)#channel-group [number] mode on

- run the above command on all interfaces you want to add to the EtherChannel

```
Switch#configure terminal
Switch(config)#interface fastethernet 0/1
Switch(config-if)#channel-group 1 mode on
Switch(config-if)#exit
Switch(config)#interface fastethernet 0/2
Switch(config-if)#channel-group 1 mode on
Switch(config-if)#exit
Switch(config)#interface fastethernet 0/3
Switch(config-if)#channel-group 1 mode on
Switch(config-if)#exit
Switch(config)#interface fastethernet 0/4
Switch(config-if)#channel-group 1 mode on
Switch(config-if)#exit
```

EtherChannel

- Use the **show spanning-tree** command again to **view the STP ports' states after etherchannel configuration.**

```
Switch#show spanning-tree
VLAN0001
  Spanning tree enabled protocol ieee
  Root ID    Priority    32769
             Address     0060.4765.73AE
             Cost        7
             Port        27 (Port-channel1)
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15

  Bridge ID  Priority    32769  (priority 32768 sys-id-ext 1)
             Address     00E0.8F8A.734B
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15
             Aging Time  20

Interface                Role Sts Cost          Prio.Nbr Type
-----
Po1                      Root FWD 7            128.27 Shr
```

EtherChannel

- View the information about specified EtherChannel

```
Switch#show etherchannel
                Channel-group listing:
                -----

Group: 1
-----
Group state = L2
Ports: 4 Maxports = 8
Port-channels: 1 Max Port-channels = 1
Protocol:      -
```

EtherChannel

- output of the show spanning-tree command on the second switch

```
Switch#show spanning-tree
VLAN0001
  Spanning tree enabled protocol ieee
  Root ID    Priority    32769
             Address     0060.4765.73AE
             This bridge is the root
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    32769  (priority 32768 sys-id-ext 1)
             Address     0060.4765.73AE
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time  20
```

Interface	Role	Sts	Cost	Prio.Nbr	Type
Fa0/2	Desg	FWD	19	128.2	P2p
Fa0/1	Desg	FWD	19	128.1	P2p
Fa0/4	Desg	FWD	19	128.4	P2p
Fa0/3	Desg	FWD	19	128.3	P2p

EtherChannel

- the configuration steps for the EtherChannel on the second switch

```
Switch#configure terminal
Switch(config)#interface fastethernet 0/1
Switch(config-if)#channel-group 1 mode on
Switch(config-if)#exit
Switch(config)#interface fastethernet 0/2
Switch(config-if)#channel-group 1 mode on
Switch(config-if)#exit
Switch(config)#interface fastethernet 0/3
Switch(config-if)#channel-group 1 mode on
Switch(config-if)#exit
Switch(config)#interface fastethernet 0/4
Switch(config-if)#channel-group 1 mode on
Switch(config-if)#exit
```

EtherChannel

- show spanning-tree command on the second switch **after the EtherChannel Configuration.**

```
Switch#show spanning-tree
VLAN0001
  Spanning tree enabled protocol ieee
  Root ID    Priority    32769
             Address     0060.4765.73AE
             This bridge is the root
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    32769  (priority 32768 sys-id-ext 1)
             Address     0060.4765.73AE
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time  20
```

Interface	Role	Sts	Cost	Prio.Nbr	Type
-----	-----	----	-----	-----	-----
Po1	Desg	FWD	7	128.27	Shr